
Narrowing Without Converging: How Interactive LLM Agents Search Under Feedback

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Abstract

1 Interactive agents should use feedback to converge on correct answers, yet how
2 they search remains poorly understood. We study three Qwen model tiers on 22
3 Turtle Soup puzzles using scored answer checkpoints and embedded trajectories
4 of successive questions. Accuracy improves for only the first 10 of 30 rounds,
5 while semantic stride contracts four- to fivefold. Longer budgets, Oracle quality,
6 and puzzle under-determination do not explain this plateau. Feedback also redi-
7 rects search poorly: after refutations, all three tiers move toward regions already
8 ruled out. The tiers also fail differently underneath the shared plateau. The small-
9 est circles a single early hypothesis; the largest reaches solution-grade stories and
10 then abandons them. Endpoint accuracy cannot distinguish these two failures. We
11 release the evaluation harness, process measures, verified puzzle set, and per-game
12 logs.

13 1 Introduction

14 Consider a Turtle Soup puzzle: *A man’s body lies in the desert, clutching half a matchstick; luggage*
15 *and clothing are scattered around him.* The solver sees only this surface and must recover the hidden
16 explanation through yes/no questions. In the solution, a hot-air balloon was losing altitude. After
17 discarding their luggage, the passengers drew lots, and the man holding the short matchstick was
18 thrown overboard.

19 Existing evaluations [1, 2, 3] can determine whether an agent eventually recovers this story, but not
20 how it searches. The same endpoint score may describe an agent that never leaves its initial hy-
21 pothesis and one that discovers the answer but later abandons it. Existing Turtle Soup benchmarks
22 emphasize outcomes rather than search dynamics (Appendix A). This leaves a central question un-
23 resolved: how does feedback change an agent’s next move?

24 Turtle Soup provides a controlled setting for answering this question. The interaction is textual, the
25 Oracle supplies a discrete yes/no/irrelevant signal, and each puzzle is deliberately under-determined.
26 We introduce two process-level probes. After every Oracle response, the Questioner produces a
27 complete checkpoint reconstruction, allowing accuracy to be tracked across rounds. We also embed
28 each question to trace the association trajectory and measure its per-round semantic stride. We
29 evaluate three Qwen model tiers on 22 human-played puzzles.

30 We test three hypotheses. **H1**: refutations progressively constrain the hypothesis space, so shorter
31 strides should accompany convergence. **H2**: feedback redirects the next question, with refutations
32 pushing the search away and confirmations promoting local refinement. **H3**: once an agent reaches
33 a solution-grade reconstruction, it preserves and commits that answer.

34 Our analysis shows that interactive search can narrow without converging. It also separates two fail-
35 ures hidden by endpoint accuracy: circling around an early hypothesis and abandoning a stronger

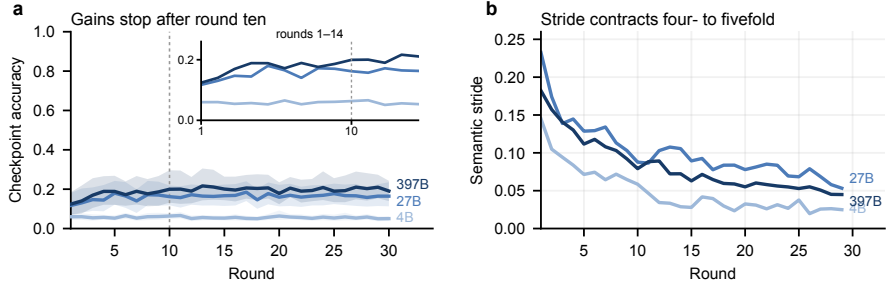


Figure 1: (a) Accuracy of the answer each tier would give at each round (bands, 95% cluster-bootstrap CIs over 22 puzzles); dashed rule at round 10; inset, rounds 1–14. (b) Mean semantic stride by round: the step between consecutive questions contracts four- to fivefold at every tier while the curves in (a) stay flat.

earlier reconstruction. Finally, we establish an important measurement boundary: embedding geometry can describe these trajectories but cannot reliably score their success. We release the evaluation harness, process instruments, verified puzzle set, and per-game logs.

2 Framework and experiments

Task and harness. A **Questioner** sees only the puzzle’s visible *surface* and asks one yes/no question per round, following the SPLAT player–judge protocol [2]. An **Oracle** holds the hidden story and answers yes, no, or irrelevant. The Questioner may submit a final story at any round and must do so when the round cap is reached the maximum number of rounds a game may run.

Checkpoint and commitment instrumentation. After every Oracle response, the Questioner writes its best complete reconstruction. We score each checkpoint, so every point on the accuracy curve represents a full answer based on the evidence available at that round. We separately record *commitment*, defined by whether and when the Questioner voluntarily submits a final story. Together, these measures convert an endpoint benchmark into a scored process.

Puzzles. To ensure that the intended explanations are demonstrably recoverable, we restrict evaluation to puzzles with a public record of human play. The resulting set contains 22 puzzles from a Chinese Turtle Soup community site, with 97 annotated key clues. Each puzzle receives an **under-determination** score: one minus the best of twelve cold guesses made from the surface alone. It predicts what a model writes before any evidence arrives. Appendix B gives the distribution and validation.

Outcome scoring. The outcome score combines **clue recall** (70 points) and **causal-logic identity** (30 points), as detailed in Appendix C. The logic component recognizes causally equivalent answers expressed through different wording. The clue component anchors answers to the annotated evidence and prevents semantically distant alternatives from receiving high scores.

Association-trajectory instrumentation. Our second instrument analyzes the questions themselves, adapting forward flow [7] and the Divergent Association Task [8] from human psychometrics. For each round, we extract keywords with jieba (a standard Chinese word segmenter), embed them using bge-small-zh-v1.5, and average the normalized embeddings into a unit vector q_t . The same extractor is used for every model, so no model is measured through its own vocabulary. All distances are cosine distances.

We use two anchors. The surface-only anchor represents terms from the visible puzzle surface, whereas the solution-aware anchor combines surface, solution, and clue terms. The primary measures are the **stride**, $s_t = d(q_t, q_{t-1})$, and the **anchor distance**, $h_t = d(q_t, \mathcal{A})$, computed separately for each anchor. H1 predicts that contraction in stride should accompany movement toward the solution and improved checkpoint accuracy. H2 is tested by pairing each Oracle response at round t with the next stride s_{t+1} and by comparing movement toward previously confirmed and refuted regions. H3 is tested by comparing *solution-grade* checkpoint peaks — those reaching 0.5 and held

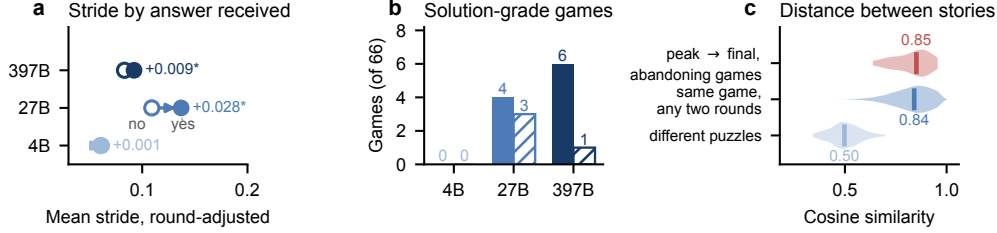


Figure 2: (a) Stride into the next question after refutation and confirmation, adjusted for round position. (b) Games reaching a solution-grade checkpoint and the subset voluntarily committed. (c) Similarity between final and peak stories in abandoning games — those whose final checkpoint falls below half their peak ($n = 34$) — relative to ordinary within-game variation (77,906 pairs from 60 games) and to stories from different puzzles (231 pairs).

across two consecutive rounds, so that a single lucky wording does not qualify — with subsequent commitment and final answers.

Experimental designs. E1 plays each game for up to 30 rounds and scores a checkpoint after every round. E2 imposes hard caps of $\{5, 10, 15, 20, 25, 30\}$ rounds and scores only the forced final answer. E3 analyzes the E1 trajectories using $\{s_t\}$, $\{h_t\}$, response-conditioned movement, and checkpoint commitment.

Model grid and validity controls. We evaluate Qwen3.5-4B, Qwen3.6-27B, and Qwen3.5-397B-A17B on 22 puzzles with three seeds. E1 therefore contains 198 thirty-round games, while E2 contains 1,188 games across the six caps. DeepSeek-V3.1 serves as both Oracle and causal-logic judge. It is cross-family with every Questioner and achieves 95% accuracy on held-out yes/no probes. Every model call uses `max_tokens=2048`, disabled thinking, temperature 0.2, and a 50,000-token per-game budget. The smallest tier sometimes exhausts its per-game token budget and is then forced to answer. We report those games separately throughout, so that a harness limit is not read as agent behaviour. The three Questioners differ in more than size: 27B is a later generation, and 397B is a mixture-of-experts with 17B active parameters. We therefore compare model tiers and do not claim a scaling law.

3 Results

Interaction improves accuracy only during the first ten rounds. Checkpoint accuracy increased through approximately round 10 and then plateaued at every tier (Figure 1a). From rounds 10 to 30, accuracy changed by -0.009 , $+0.003$, and -0.013 for 4B, 27B, and 397B, respectively; all confidence intervals included zero. Semantic stride contracted monotonically over the same rounds (Figure 1b): the search narrowed while accuracy did not move.

The plateau was not explained by the round cap. Final accuracy was insensitive to it for the two larger tiers, and the apparent decline at 4B tracked token exhaustion (0/66 to 63/66 games across caps) rather than the cap itself. The Oracle answered 95% of held-out probes correctly, with 68–74% refusals, so the feedback channel had not collapsed. Nor did puzzle under-determination explain it: improvement after round 10 was unrelated to how open a puzzle is, at every tier ($|\rho| \leq 0.10$, n.s. at $n = 22$).

H1: does the search narrow toward the answer? It narrows, but not toward the answer. Mean stride decreased monotonically at every tier, falling four- to fivefold from 0.145/0.232/0.182 at round 1 to 0.025/0.053/0.045 at round 29. This contraction did not accompany convergence: within-tier stride did not predict final performance ($|\rho| \leq 0.16$). The 4B tier had the smallest late-game stride, 0.030 [0.021, 0.039], compared with 0.082 [0.065, 0.098] for 27B, yet never reached a solution-grade peak. Its trajectory therefore contracted around a persistent hypothesis rather than toward the solution, the pattern we call *circling*.

H2: does the answer redirect the next question? Not in the direction predicted, and at the smallest tier not at all. We paired each Oracle response with the next stride and controlled for round position within puzzle and round-bin cells. Let Δs denote stride after confirmation minus stride after refutation. No difference was detected for 4B (+0.001 [−0.009, +0.011]), whereas 27B and 397B moved farther after confirmation (+0.028 [+0.010, +0.046] and +0.009 [+0.002, +0.017]; Figure 2a). Response-conditioned movement was therefore absent at the smallest tier and opposite to the hypothesized direction at the larger tiers. The control matters: confirmations arrive earlier than refutations and early strides are larger, so without it an apparent effect appears at every tier (Appendix C reports both estimates).

A confirmation is the rarer answer and could reasonably justify a larger step, so we also ran a test that assumes nothing about the right step size: after each response, is the next question nearer to regions the Oracle has already refuted, or to those it has already confirmed? After a refutation every tier moved toward the refuted side: −0.089 [−0.112, −0.068] for 4B, −0.051 [−0.075, −0.032] for 27B, and −0.032 [−0.047, −0.018] for 397B. After a confirmation no such asymmetry appeared, although refutations outnumber confirmations roughly three to one in both comparisons, so the effect is not an artifact of the larger set. Feedback therefore failed to steer any tier away from rejected regions.

H3: does the agent stop when it finds the answer? Rarely, and least often at the tier that found the most. The 4B tier reached no solution-grade checkpoint. The 27B tier reached solution-grade checkpoints in four games and voluntarily committed in three; 397B did so in six games but committed in only one (Figure 2b). Commitment grew rarer and later with scale: 11/66, 9/66, and 2/66 games at mean rounds 11.1, 14.2, and 23.5 for 4B, 27B, and 397B. The largest tier therefore reached useful hypotheses most often and preserved them least. The next paragraph measures that failure on a looser threshold that admits near-misses: *abandoning* games are those whose best checkpoint reaches 0.3 and whose final falls below half of it ($n = 34$).

Embedding geometry describes abandonment but does not score it. In abandoning games, the median cosine similarity between the final and peak stories was 0.851. This was comparable to the similarity between checkpoints from the same game (0.841) and much greater than that between stories from different puzzles (0.498; Figure 2c). Despite this semantic proximity, the median score across the same peak-to-final pairs falls from 0.447 to 0.099. The decisive substitution was therefore smaller than the encoder’s resolution, which is why the trajectory geometry characterized how agents failed without predicting how well they scored (Appendix C).

4 Limitations and Conclusion

Limitations. Our evidence comes from one game, one Oracle protocol, and three Qwen-family Questioners. The trajectory measures capture how far each question moves in embedding space, not what operation it performs; round position is controlled only by binning, the anchors are not human-derived, and the encoder’s resolution is itself a limit on what any embedding-based instrument can score (§3). Token exhaustion can mimic inability to form questions; the Oracle’s 95% audited accuracy implies roughly one wrong response per game; and using the same DeepSeek model as Oracle and logic judge may correlate their errors. Under-determination may partly measure puzzle familiarity, though no claim rests on it. The puzzles include death and other dark themes and carry content flags.

Conclusion. The agents narrow without converging: semantic stride decreases four- to fivefold while accuracy plateaus, and feedback fails to steer questions consistently away from rejected regions. The smallest tier mostly circles early hypotheses, whereas the largest tier sometimes reaches solution-grade reconstructions but fails to preserve or commit them. The two failures require different fixes yet appear identical under endpoint accuracy; process-level measurement reveals how search fails even when it cannot predict performance. Whether these patterns extend to other tasks, model families, and protocols remains open.

Declaration of LLM usage

LLMs are the object of study: the Questioner agents, the Oracle, and the causal-logic judge are all LLMs, specified in §2 and Appendix C, and every reported number derives from their logged interactions. Separately, LLM assistants were used to help write analysis code and edit prose; the authors reviewed all analyses, claims, and text, and take full responsibility for them.

References

- [1] Yu, Q., Song, S., Fang, K., Shi, Y., Zheng, Z., Wang, H., et al. (2024) TurtleBench: Evaluating top language models via real-world yes/no puzzles. arXiv:2410.05262.
- [2] Chen, Q., Zhang, B., Wang, G., & Wu, Q. (2024) Weak-eval-Strong: Evaluating and eliciting lateral thinking of LLMs with situation puzzles. arXiv:2410.06733.
- [3] Zhou, M., Wu, S., Zhang, H., Sima, Q., & Liu, B. (2025) What to ask next? Probing the imaginative reasoning of LLMs with TurtleSoup puzzles. arXiv:2508.10358.
- [4] Park, J., Baek, I., Park, J., & Lee, H. (2026) Beyond one path: Evaluating and enhancing divergent thinking in interactive LLM agents. arXiv:2605.28465.
- [5] Ye, J., Wang, Y., Huang, Y., Chen, D., Zhang, Q., Moniz, N., et al. (2025) Justice or prejudice? Quantifying biases in LLM-as-a-Judge. In *International Conference on Learning Representations (ICLR)*. arXiv:2410.02736.
- [6] Shi, L., Ma, C., Liang, W., Diao, X., Ma, W., & Vosoughi, S. (2025) Judging the judges: A systematic study of position bias in LLM-as-a-Judge. In *Proceedings of IJCNLP–AAACL 2025*, pp. 292–314. arXiv:2406.07791.
- [7] Gray, K., Anderson, S., Chen, E. E., Kelly, J. M., Christian, M. S., Patrick, J., et al. (2019) “Forward flow”: A new measure to quantify free thought and predict creativity. *American Psychologist* 74(5):539–554.
- [8] Olson, J. A., Nahas, J., Chmoulevitch, D., Cropper, S. J., & Webb, M. E. (2021) Naming unrelated words predicts creativity. *Proceedings of the National Academy of Sciences* 118(25):e2022340118.
- [9] De Deyne, S., Navarro, D. J., Perfors, A., Brysbaert, M., & Storms, G. (2019) The “Small World of Words” English word association norms for over 12,000 cue words. *Behavior Research Methods* 51(3):987–1006.

A Related work

Existing Turtle Soup benchmarks score outcomes. TurtleBench [1], SPLAT [2] (whose interactive player-judge protocol we adopt), and TurtleSoup-Bench [3] evaluate verification accuracy, solve efficiency, or faithfulness to the intended story. TurtleSoup-Bench also probes what agents ask next, but it grades the question against the intended story; none of the three measures the geometry of the search itself, which is the gap this paper addresses. MUTATE [4] shows that frontier agents find far fewer mechanism-distinct solution paths than humans; its diverge-then-narrow agent is a natural intervention to test with our instruments. The trajectory instrument adapts forward flow [7] and the Divergent Association Task [8] from human psychometrics, and the Small World of Words norms [9] are the natural source for the human-derived anchor that ours is not yet (§2). Catalogues of judge bias [5, 6] motivated confining the model judge to the one subscore that objective matching cannot reach (Appendix C).

B The puzzle set

Selection. The 22 puzzles were hand-picked from the classic repertoire of a Chinese Turtle Soup community, restricted to puzzles with a public record of human play (§2): a puzzle nobody has solved carries no evidence that its clues suffice or that the intended leap is reachable. Every candidate was read and kept or rejected by hand. Surfaces run 9–108 characters (median 34) and solutions 13–200 characters (median 102).

Distribution of under-determination. Under-determination is defined in §2. Its distribution over the 22 puzzles is:

Under-determination	Puzzles
0.15–0.25	5
0.25–0.35	8
0.35–0.45	4
0.45–0.55	4
0.55–0.65	1

The median is 0.347, the quartiles are 0.273 and 0.410, and the full range is 0.173–0.577: single-peaked with a thin hard tail. The easiest puzzles are the story the genre is named for (0.17) and the desert matchstick (0.19); the hardest is a fourteen-character surface reading *our heights differ / a flowerpot broke / our heights are the same* (0.58).

Validity as a covariate. The score describes the puzzle, not any agent’s performance on it, and it is independent of rounds used; a difficulty defined as “how many rounds this takes” would make an accuracy-versus-rounds plot confirm itself. It is not a proxy for surface length ($r = -0.22$, n.s. at $n = 22$, a weak test). It is also one measure rather than two: a surface loose enough to admit many mechanisms is, for the same reason, one whose true mechanism takes more reframing to reach, so *how far* and *how many ways* are not separable here.

Familiarity caveat. The two puzzles scoring easiest are among the most widely circulated, and the models’ first-round guesses are consistent with familiarity contributing: on the best-known of them, the two larger tiers open at 0.61 and 0.53 before any evidence arrives, against 0.24 and 0.36 on a puzzle of almost identical under-determination. On the easy end the measure may therefore read familiarity as much as difficulty. Separating the two would take a memorisation probe, which we have not run; no claim in the paper rests on the measure.

C Measuring agent performance

Outcome score. The outcome score combines **clue recall (70 points)** and **causal-logic identity (30 points)** (§2). Clue recall matches the answer against per-puzzle annotated key clues and is objective and reproducible; matching tolerates paraphrase through character-bigram recall, which sets a floor on how short an annotated clue may be (below the floor only a verbatim match counts,

so all clues are annotated above it). The logic judge rates only whether cause $\rightarrow$ mechanism $\rightarrow$ outcome matches and is instructed to ignore wording. Single ratings are unstable on borderline answers, so each rating is sampled twice at temperature 0.2 and averaged, which is enough to make the judged half reproducible: the two samples agree exactly in 86% of ratings and differ by under one point of 30 on average.

Causal logic needs its own score. Models reach the right causal shape before, and often without, landing the annotated clue content. Every tier therefore earns proportionally far more of the judged causal-logic half than of the clue-recall half — three- to fourfold at 397B (41% against 10%) and 27B (33% against 10%), and thirtyfold at 4B, which earns 0.5% of the clue half — and the halves correlate only weakly per game (Spearman $\rho = +0.12 / +0.40 / -0.21$ for 397B / 27B / 4B over the 66 final answers per tier). Scored on clue recall alone, the smallest tier would sit at the floor and the other two would read as near-total failures. The 70/30 weighting is a design choice.

Controlling for round position removes the smallest tier’s response. Uncontrolled, every tier appears to answer feedback; controlled, the smallest does not. For each round t the Oracle’s answer is paired with s_{t+1} , the stride into the next question; “irrelevant” rounds are excluded. Confirmation arrives earlier than refutation (mean round 12.1 / 9.7 / 10.9 for a “yes” against 15.6 / 15.6 / 15.9 for a “no”) and early strides are larger, so an uncontrolled comparison is confounded by position. Pairs are formed inside (puzzle, round-bin) cells with bins {1–5, 6–10, 11–15, 16–20, 21–25, 26–29}, and 95% CIs bootstrap over puzzles (4,000 resamples). Uncontrolled, all three tiers appear to respond (+0.024 [+0.004, +0.044]; +0.043 [+0.015, +0.070]; +0.034 [+0.023, +0.046]); controlled, the smallest does not (+0.001 [−0.009, +0.011]; +0.028 [+0.010, +0.046]; +0.009 [+0.002, +0.017]).

Search behaviour does not track puzzle structure. How open a puzzle is changes what an agent knows at round 1, but not how it searches afterwards. Regressing each per-trace indicator on under-determination, seeds averaged, 5,000 permutations: stride, late stride, per-round change in distance to each anchor, mean anchor distance and rounds played are all n.s. at every tier ($|\rho| \leq 0.32$, $p > 0.10$). Commit rate is the one indicator that leans: negative at all three tiers (−0.32 / −0.25 / −0.43), so more open puzzles draw fewer commitments. But pooled $p = 0.13$, and 1 of 24 tests reaching $p < 0.05$ is the chance expectation, so we report it as unresolved rather than as an effect.

Trajectory geometry does not predict the score. Every trajectory measure is flat against accuracy, and the intervals are narrow enough to bound the effect rather than merely fail to find one. Best checkpoint accuracy per game; Gaussian LMM, puzzle random intercept (the same 22 puzzles are replayed by every tier, so games are not independent), tier fixed effect, ML fits, likelihood-ratio tests against the covariate-only model. Stride $\beta = -0.010$ per SD, 95% CI [−0.035, 0.014], $p = 0.42$; per-round change in distance to the solution-aware anchor $\beta = -0.011$, [−0.035, 0.014], $p = 0.39$; stride $\times$ tier $p = 0.20$; the same for the surface-only anchor $\beta = -0.020$, [−0.044, 0.003], $p = 0.097$, the closest any measure comes and still not claimed. Figure 3 shows the underlying per-trace geometry.

The Oracle passed a pre-deployment audit. The deployed DeepSeek-V3.1 scored 20/21 (95%) on yes/no probes and 3/4 exact on “irrelevant”; candidates below the 90% bar on the yes/no items, or from the Questioners’ own family, were rejected before any agent was run. The audit probes a candidate with 25 hand-crafted (question, expected-answer) pairs over three puzzles, 21 with a definite yes/no answer and 4 expecting “irrelevant”. Model-call settings are given in §2.

Game budget. Exhausting the 50,000-token game budget forces a final answer and is recorded as the game outcome `token_budget`.

Trajectory and difficulty pipeline. The under-determination covariate is scored independently of the outcome apparatus: it prompts GLM-5.3-flash (cross-family with all Questioners) for the twelve cold guesses and scores closeness as embedding similarity between guess and solution under the same encoder, not with the composite score. The embedding pipeline is specified in §2; two further details matter for reproduction. Keyword extraction strips Latin characters and digits before jieba segmentation, and late-game stride denotes the second half of a game’s strides.

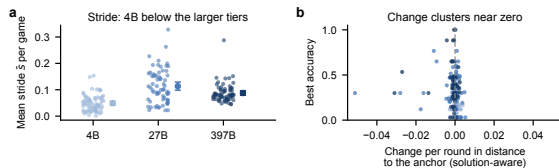


Figure 3: Per-trace geometry ($n = 198$). **(a)** Per-game mean stride by tier (dots; bars, tier mean $\pm$ 95% CI): 4B sits below the larger tiers. **(b)** Per-round change in a game’s distance to the anchor, against best accuracy: clustered near zero at every tier.

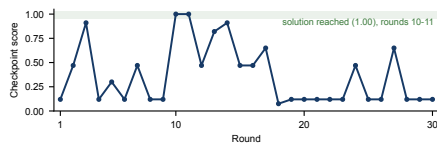


Figure 4: One abandoning game (refsoup_021, 397B, seed 2): the round 10–11 checkpoint stories are the solution (band), never volunteered, abandoned by round 19. Table 1 shows seven of the thirty rounds.

Artifacts. The harness, the puzzle set with per-item source records, per-game logs including every checkpoint story, and the analysis script behind each reported number are released in a public repository, withheld here for double-blind review. Puzzle text comes from a public community repertoire and is redistributed with attribution and a content flag.

One game, worked. Figure 4 and Table 1 show puzzle refsoup_021 (surface: a stormy night, a run-down temple, two photographs on the wall that seem to stare; by morning the narrator sees only two windows; solution: “there were never photos on the wall; people were pressing against the windows, looking in”), played by Qwen3.5-397B, seed 2. Questions and stories are our translations of the logged Chinese, with stories abridged to the clause carrying the hypothesis. Rounds 22–30, eight re-wordings of the same flooding hypothesis (e.g. round 25, whether the narrator woke floating upside down; round 28, whether rising water reached eye level), are in the released logs.

Rd	Question (abridged)	Ans.	Committed story (abridged)	score (clue+logic)
1	Are the ones in the photos people?	yes	The two “photos” are two people nailed or sealed into wall niches.	0.12 (0+12)
3	On waking the next day, is he still in the temple?	yes	The “two photos” are really two windows, mistaken for photos in the dim light.	0.91 (70+21)
4	Are the “two windows” the same objects as last night’s “photos”?	yes	The “photos” were hollowed-out window openings; the fear came from darkness and illusion.	0.12 (0+12)
10	Did frame glass break and fall, leaving open holes?	no	They were never photos, but faces pressed against the outside of the windows.	1.00 (70+30)
11	Were the photos removed, revealing windows behind?	no	The staring figures were people outside the windows.	1.00 (70+30)
19	Were real people hiding behind the photos?	no	Window openings sealed by painted cloth covers; the storm tore them off.	0.12 (0+12)
30	Did rising water lift the narrator higher?	no	Papered-over window openings; the storm stripped the paper away.	0.12 (0+12)

Table 1: Seven rounds of one E1 game (refsoup_021, Qwen3.5-397B, seed 2): the Questioner’s question, the Oracle’s answer, the story the agent was forced to commit that round, and its composite score. Questions and stories are our translations; the original transcripts are in the released logs.